

Sri Lanka Institute of Information Technology

IE3132 – Penetration Testing

Assignment 02 - CTF Play Box Implementation – Working Box Demonstration

Project Name – Kernal0X

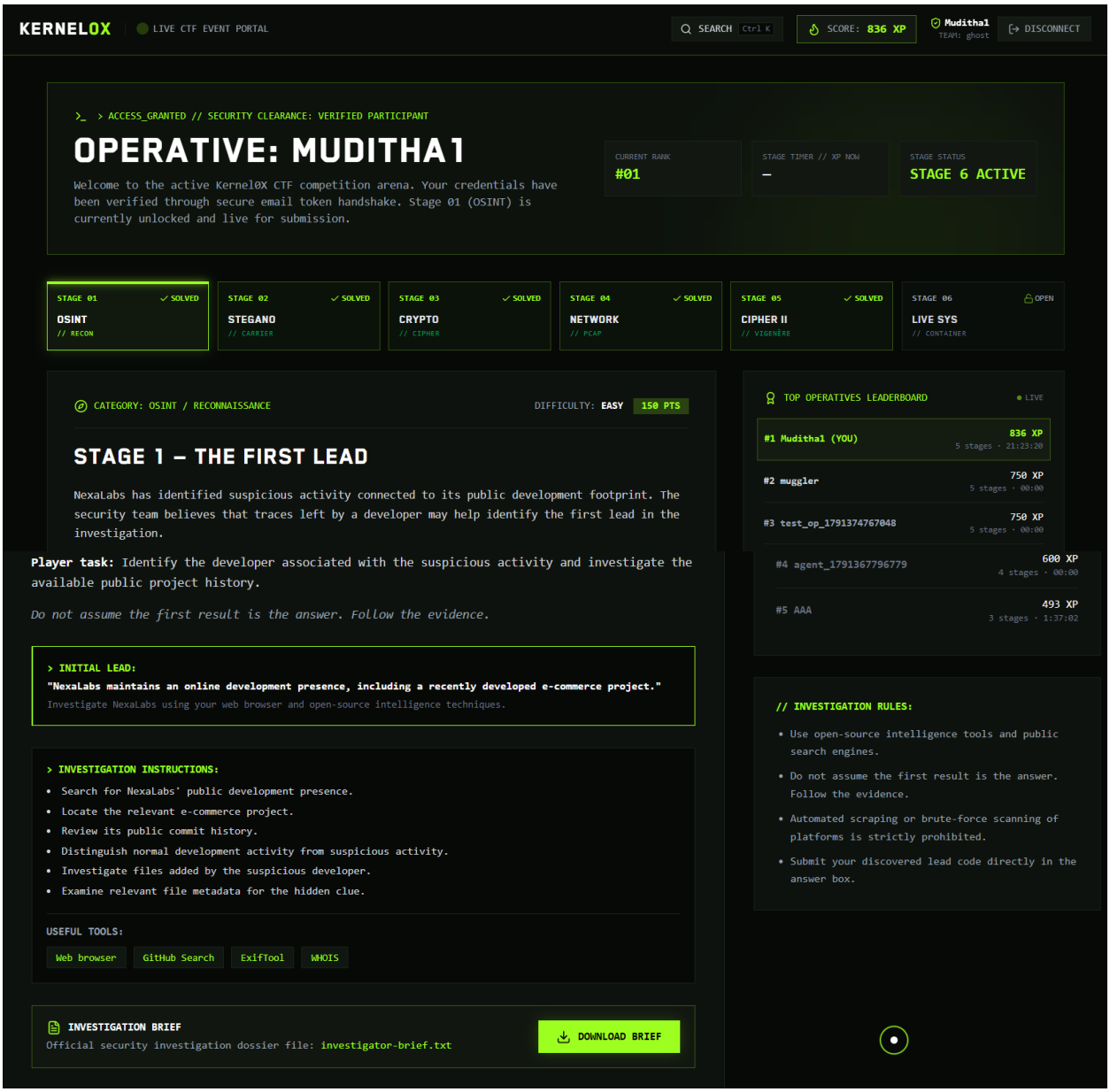
Group Members

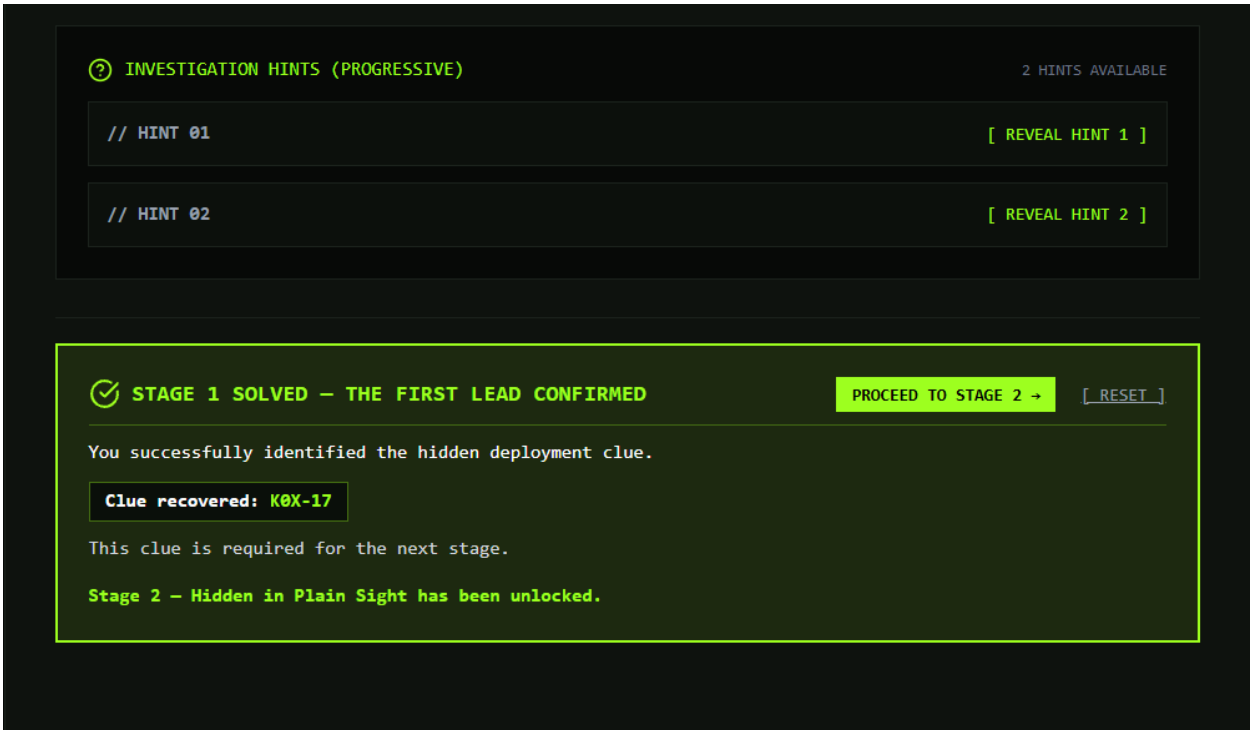
Student Name	Registration No.
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Evidence

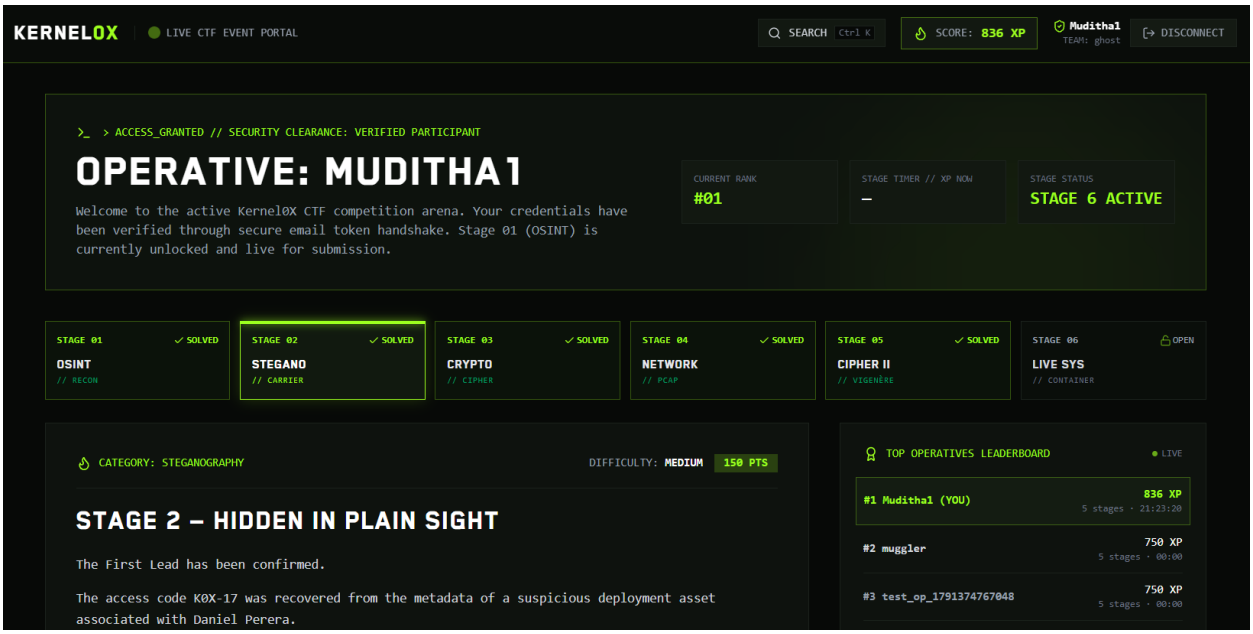
Completed stages

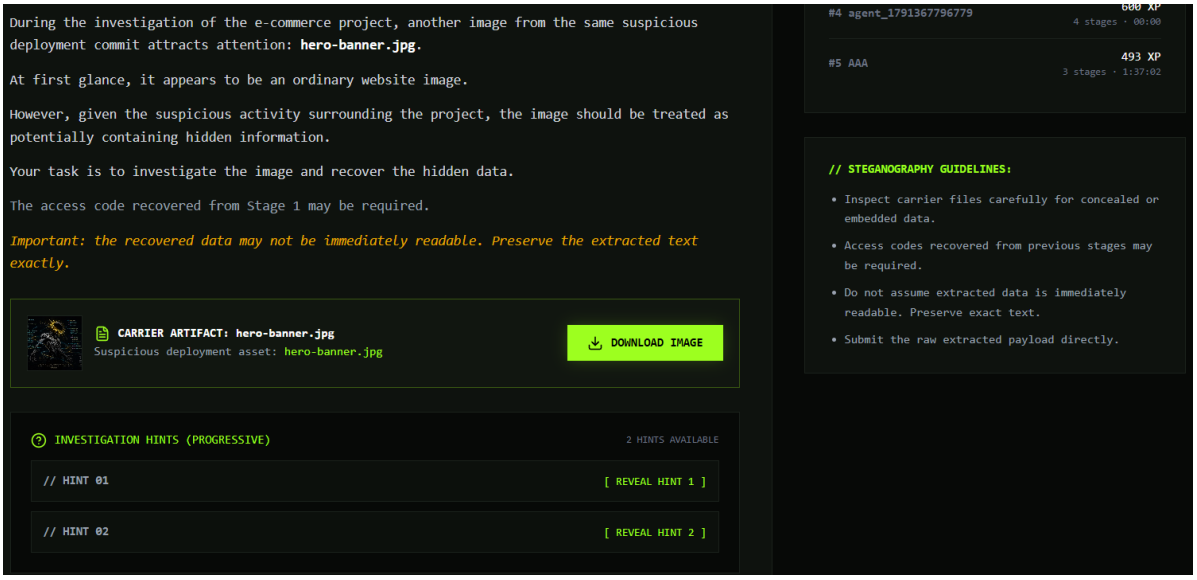
Stage 1

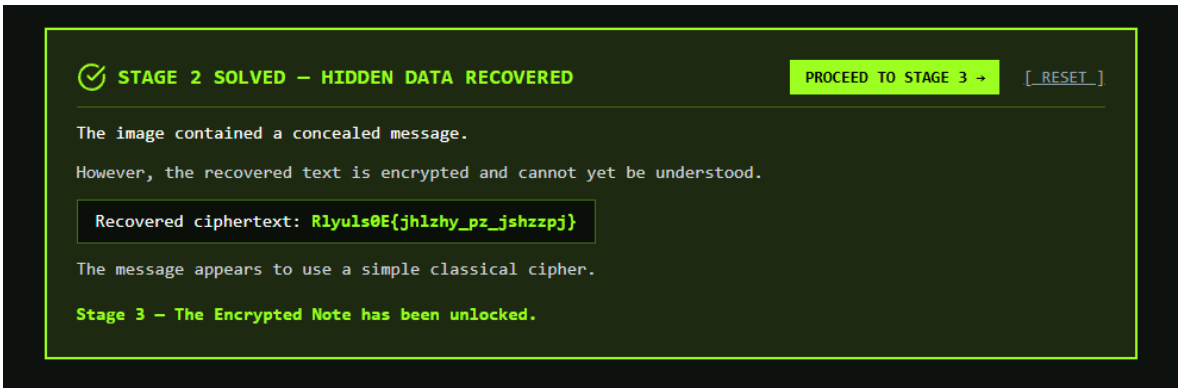



INVESTIGATION HINTS (PROGRESSIVE)
2 HINTS AVAILABLE
// HINT 01
[REVEAL HINT 1]
// HINT 02
[REVEAL HINT 2]
STAGE 1 SOLVED — THE FIRST LEAD CONFIRMED
PROCEED TO STAGE 2 →
[RESET]
You successfully identified the hidden deployment clue.
Clue recovered: K0X-17
This clue is required for the next stage.
Stage 2 — Hidden in Plain Sight has been unlocked.

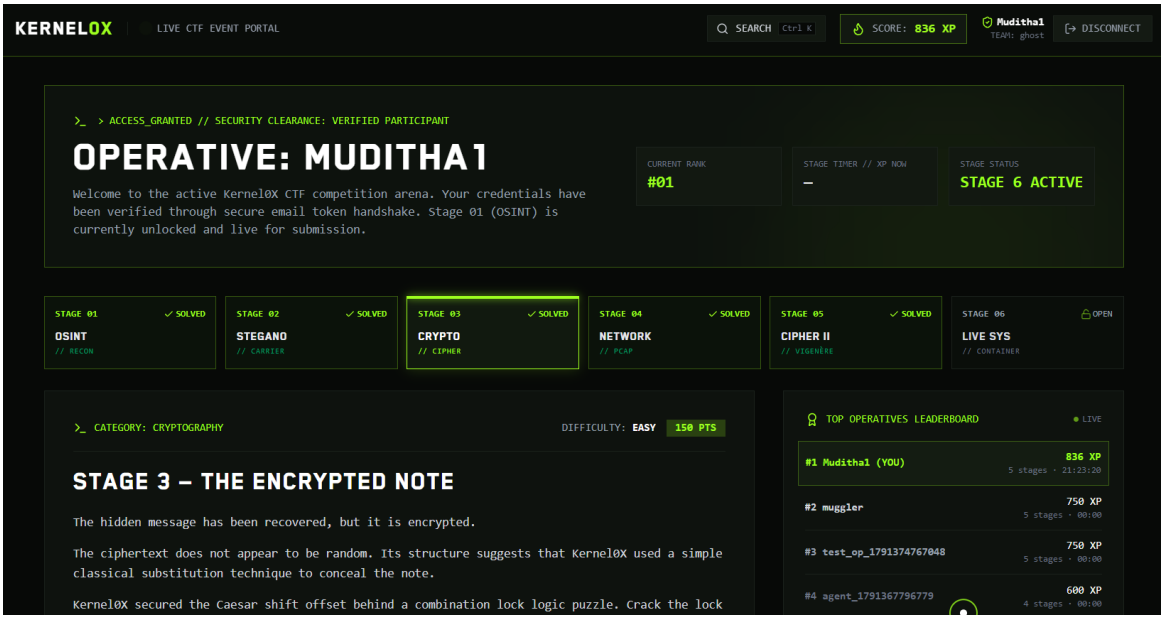
Stage 2


KERNELOX
LIVE CTF EVENT PORTAL
SEARCH Ctrl K
SCORE: 836 XP
Muditha1
TEAM: ghost
DISCONNECT
> ACCESS_GRANTED // SECURITY CLEARANCE: VERIFIED PARTICIPANT
OPERATIVE: MUDITHA1
Welcome to the active Kernel0X CTF competition arena. Your credentials have been verified through secure email token handshake. Stage 01 (OSINT) is currently unlocked and live for submission.
CURRENT RANK
#01
STAGE TIMER // XP NOW
—
STAGE STATUS
STAGE 6 ACTIVE
STAGE 01 SOLVED
OSINT
// RECON
STAGE 02 SOLVED
STEGANO
// CARRIER
STAGE 03 SOLVED
CRYPTO
// CIPHER
STAGE 04 SOLVED
NETWORK
// PCAP
STAGE 05 SOLVED
CIPHER II
// VIGENÈRE
STAGE 06 OPEN
LIVE SYS
// CONTAINER
CATEGORY: STEGANOGRAPHY
DIFFICULTY: MEDIUM
150 PTS
STAGE 2 — HIDDEN IN PLAIN SIGHT
The First Lead has been confirmed.
The access code K0X-17 was recovered from the metadata of a suspicious deployment asset associated with Daniel Perera.
TOP OPERATIVES LEADERBOARD
LIVE
#1 Muditha1 (YOU)
836 XP
5 stages · 21:23:20
#2 muggler
750 XP
5 stages · 00:00
#3 test_op_1791374767048
750 XP
5 stages · 00:00


During the investigation of the e-commerce project, another image from the same suspicious deployment commit attracts attention: hero-banner.jpg.
At first glance, it appears to be an ordinary website image.
However, given the suspicious activity surrounding the project, the image should be treated as potentially containing hidden information.
Your task is to investigate the image and recover the hidden data.
The access code recovered from Stage 1 may be required.
Important: the recovered data may not be immediately readable. Preserve the extracted text exactly.
CARRIER ARTIFACT: hero-banner.jpg
Suspicious deployment asset: hero-banner.jpg
DOWNLOAD IMAGE
INVESTIGATION HINTS (PROGRESSIVE)
2 HINTS AVAILABLE
// HINT 01
[REVEAL HINT 1]
// HINT 02
[REVEAL HINT 2]
#4 agent_1791367796779
600 XP
4 stages · 00:00
#5 AAA
493 XP
3 stages · 1:37:02
// STEGANOGRAPHY GUIDELINES:
Inspect carrier files carefully for concealed or embedded data.
Access codes recovered from previous stages may be required.
Do not assume extracted data is immediately readable. Preserve exact text.
Submit the raw extracted payload directly.


STAGE 2 SOLVED — HIDDEN DATA RECOVERED
PROCEED TO STAGE 3 →
[RESET]
The image contained a concealed message.
However, the recovered text is encrypted and cannot yet be understood.
Recovered ciphertext: R1yu1s0E{jh1zhy_pz_jshzzpj}
The message appears to use a simple classical cipher.
Stage 3 — The Encrypted Note has been unlocked.

Stage 3


KERNELOX
LIVE CTF EVENT PORTAL
SEARCH Ctrl K
SCORE: 836 XP
Muditha1
TEAM: ghost
DISCONNECT
> ACCESS_GRANTED // SECURITY CLEARANCE: VERIFIED PARTICIPANT
OPERATIVE: MUDITHA1
Welcome to the active Kernel0X CTF competition arena. Your credentials have been verified through secure email token handshake. Stage 01 (OSINT) is currently unlocked and live for submission.
CURRENT RANK
#01
STAGE TIMER // XP NOW
—
STAGE STATUS
STAGE 6 ACTIVE
STAGE 01 SOLVED OSINT // RECON
STAGE 02 SOLVED STEGANO // CARRIER
STAGE 03 SOLVED CRYPTO // CIPHER
STAGE 04 SOLVED NETWORK // PCAP
STAGE 05 SOLVED CIPHER II // VIGENÈRE
STAGE 06 OPEN LIVE SYS // CONTAINER
> CATEGORY: CRYPTOGRAPHY
DIFFICULTY: EASY
150 PTS
STAGE 3 — THE ENCRYPTED NOTE
The hidden message has been recovered, but it is encrypted.
The ciphertext does not appear to be random. Its structure suggests that Kernel0X used a simple classical substitution technique to conceal the note.
Kernel0X secured the Caesar shift offset behind a combination lock logic puzzle. Crack the lock
TOP OPERATIVES LEADERBOARD
LIVE
#1 Muditha1 (YOU) 836 XP 5 stages · 21:23:20
#2 muggler 750 XP 5 stages · 00:00
#3 test_op_1791374767048 750 XP 5 stages · 00:00
#4 agent_1791367796779 600 XP 4 stages · 00:00

> RECOVERED CIPHERTEXT (STAGE 2 PAYLOAD):

```
Rlyuls0E{jhlzhy_pz_jshzzpj}
```

No new file is required for this stage. Decode this recovered ciphertext using the 1-digit shift key recovered from the lock puzzle below.

// CRACK THE LOCK — FIND THE CAESAR SHIFT (1 DIGIT)

LOCK DEDUCTION PUZZLE

To uncover the Caesar cipher shift offset, Kernel0X left this combination lock deduction puzzle. Analyze the five clues below to deduce the secret **1-digit** shift:

LOCK CONSTRAINT MATRIX: ANSWER IS 1 DIGIT

2 4 6 5	– **2 digits correct**, all wrongly placed
4 7 3 8	– **2 digits correct**, all wrongly placed
9 2 5 4	– **2 digits correct**, all wrongly placed
7 8 4 6	› **1 digit correct, rightly placed**
5 6 4 9	– **1 digit correct**, wrongly placed

TUMBLER REGISTER: ? - - -

// CRYPTOGRAPHY GUIDELINES:

- Identify the cipher type and substitution offset.
- No additional files are required for this stage.
- External tools like CyberChef or Python scripts may be used.
- Submit the decrypted flag using standard Kernel0X{...} format.

TEST YOUR 1-DIGIT SHIFT KEY GUESS:

ENTER 1 DIGIT (0-9)...

STAGE 3 SOLVED — THE ENCRYPTED NOTE HAS BEEN DECODED

PROCEED TO STAGE 4 →

[RESET]

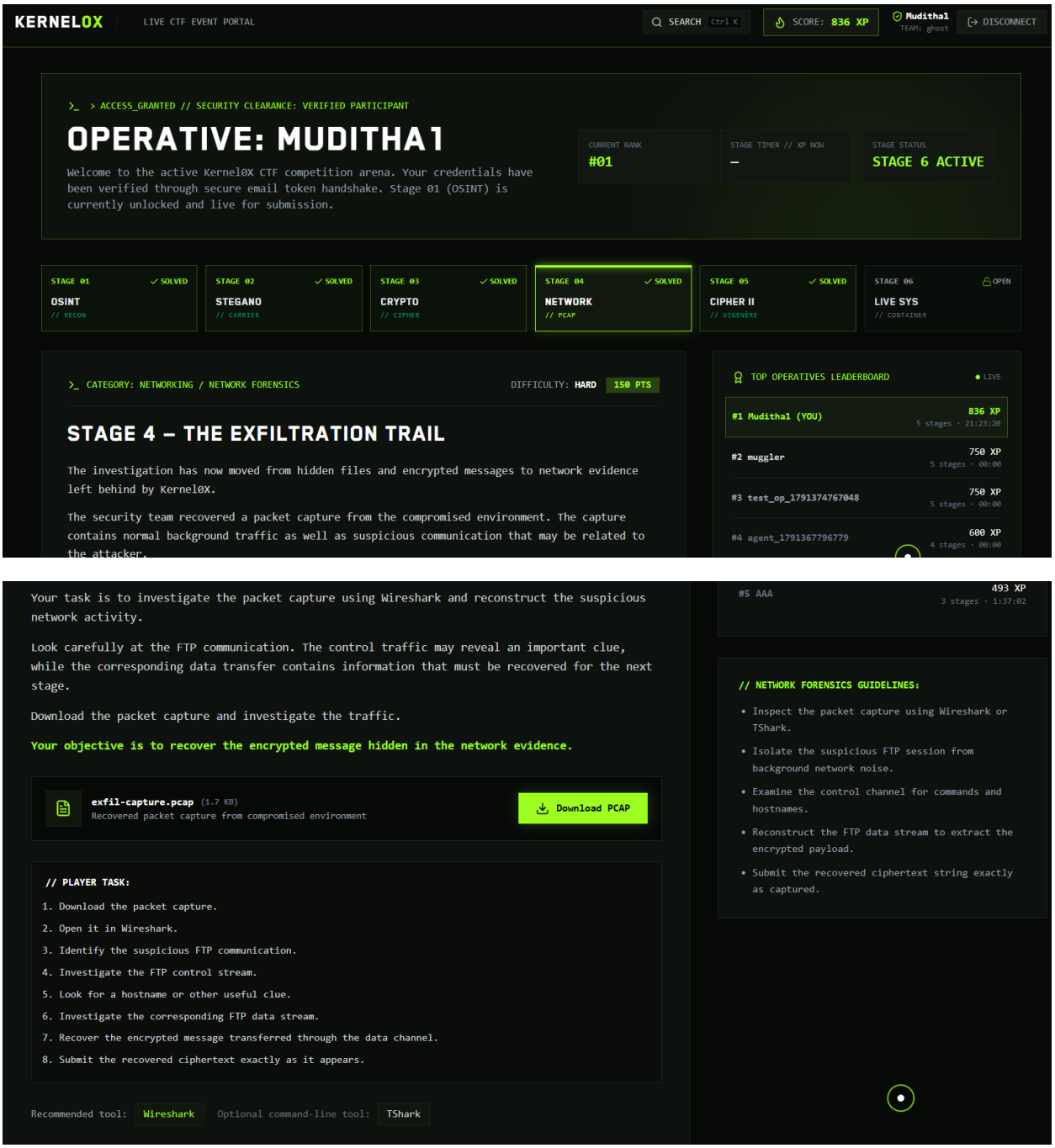
The encrypted note has been successfully decoded.

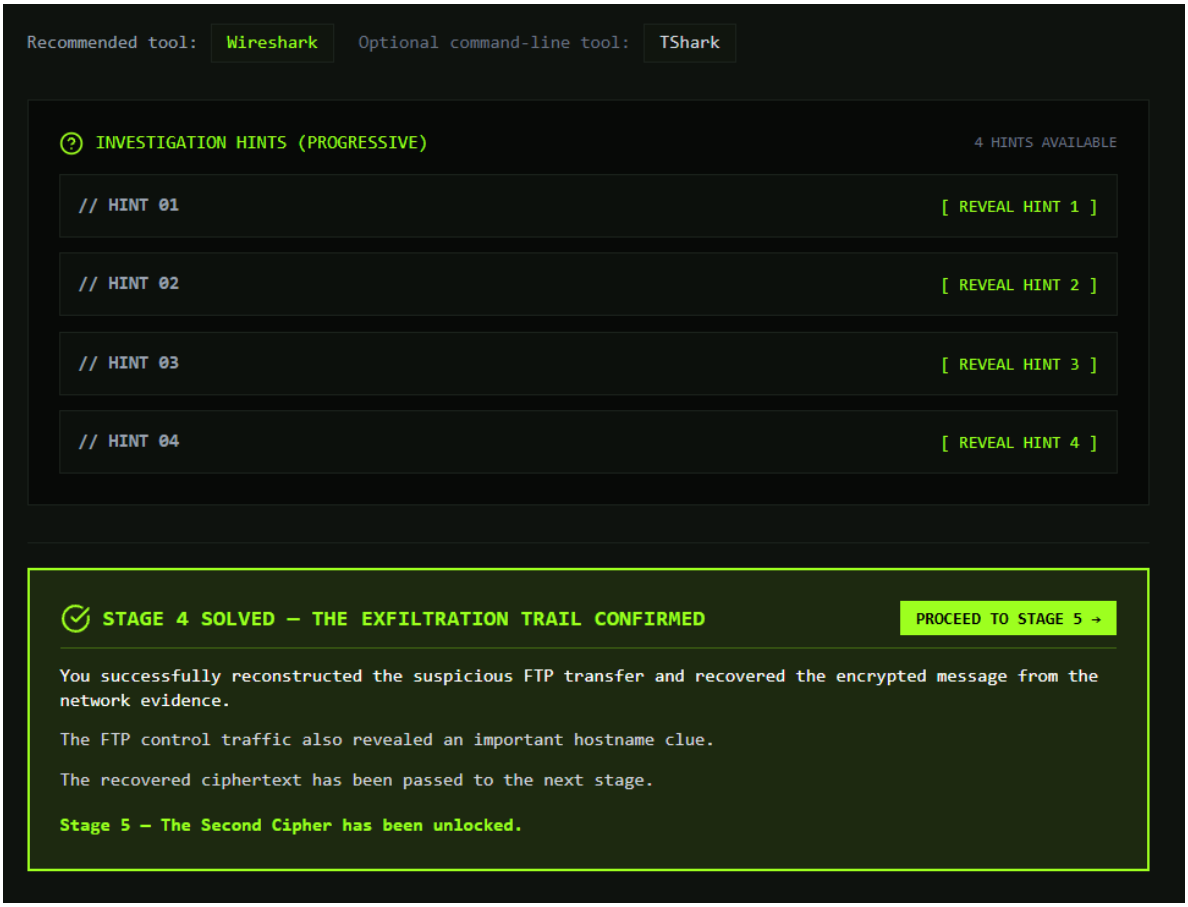
Flag recovered: **Kernel0X{caesar_is_classic}**

The investigation now moves from hidden files and encrypted messages to the network evidence left behind by the attacker.

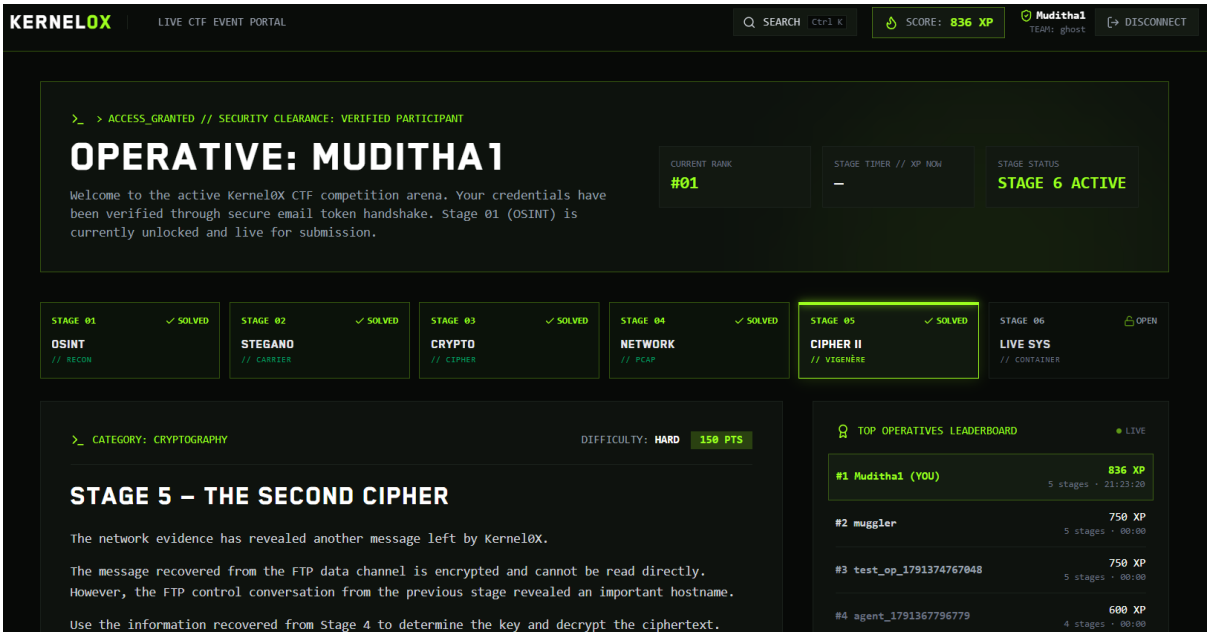
Continue to Stage 4 — The Exfiltration Trail.

Stage 4


KERNEL0X
LIVE CTF EVENT PORTAL
SEARCH Ctrl K
SCORE: 836 XP
Muditha1
TEAM: ghost
DISCONNECT
> ACCESS_GRANTED // SECURITY CLEARANCE: VERIFIED PARTICIPANT
OPERATIVE: MUDITHA1
Welcome to the active Kernel0X CTF competition arena. Your credentials have been verified through secure email token handshake. Stage 01 (OSINT) is currently unlocked and live for submission.
CURRENT RANK
#01
STAGE TIMER // XP NOW
—
STAGE STATUS
STAGE 6 ACTIVE
STAGE 01 SOLVED
OSINT
// RECON
STAGE 02 SOLVED
STEGANO
// CARRIER
STAGE 03 SOLVED
CRYPTO
// CIPHER
STAGE 04 SOLVED
NETWORK
// PCAP
STAGE 05 SOLVED
CIPHER II
// VIGENÈRE
STAGE 06 OPEN
LIVE SYS
// CONTAINER
> CATEGORY: NETWORKING / NETWORK FORENSICS
DIFFICULTY: HARD
150 PTS
STAGE 4 — THE EXFILTRATION TRAIL
The investigation has now moved from hidden files and encrypted messages to network evidence left behind by Kernel0X.
The security team recovered a packet capture from the compromised environment. The capture contains normal background traffic as well as suspicious communication that may be related to the attacker.
TOP OPERATIVES LEADERBOARD
LIVE
#1 Muditha1 (YOU) 836 XP 5 stages · 21:23:20
#2 muggler 750 XP 5 stages · 00:00
#3 test_op_1791374767048 750 XP 5 stages · 00:00
#4 agent_1791367796779 600 XP 4 stages · 00:00
#5 AAA 493 XP 3 stages · 1:37:02
Your task is to investigate the packet capture using Wireshark and reconstruct the suspicious network activity.
Look carefully at the FTP communication. The control traffic may reveal an important clue, while the corresponding data transfer contains information that must be recovered for the next stage.
Download the packet capture and investigate the traffic.
Your objective is to recover the encrypted message hidden in the network evidence.
exfil-capture.pcap (1.7 KB)
Recovered packet capture from compromised environment
Download PCAP
// PLAYER TASK:
1. Download the packet capture.
2. Open it in Wireshark.
3. Identify the suspicious FTP communication.
4. Investigate the FTP control stream.
5. Look for a hostname or other useful clue.
6. Investigate the corresponding FTP data stream.
7. Recover the encrypted message transferred through the data channel.
8. Submit the recovered ciphertext exactly as it appears.
Recommended tool: Wireshark
Optional command-line tool: TShark
// NETWORK FORENSICS GUIDELINES:
• Inspect the packet capture using Wireshark or TShark.
• Isolate the suspicious FTP session from background network noise.
• Examine the control channel for commands and hostnames.
• Reconstruct the FTP data stream to extract the encrypted payload.
• Submit the recovered ciphertext string exactly as captured.


Recommended tool: Wireshark
Optional command-line tool: TShark
INVESTIGATION HINTS (PROGRESSIVE)
4 HINTS AVAILABLE
// HINT 01
[REVEAL HINT 1]
// HINT 02
[REVEAL HINT 2]
// HINT 03
[REVEAL HINT 3]
// HINT 04
[REVEAL HINT 4]
STAGE 4 SOLVED — THE EXFILTRATION TRAIL CONFIRMED
PROCEED TO STAGE 5 →
You successfully reconstructed the suspicious FTP transfer and recovered the encrypted message from the network evidence.
The FTP control traffic also revealed an important hostname clue.
The recovered ciphertext has been passed to the next stage.
Stage 5 — The Second Cipher has been unlocked.

Stage 5


KERNEL0X
LIVE CTF EVENT PORTAL
SEARCH Ctrl K
SCORE: 836 XP
Mudithal
TEAM: ghost
DISCONNECT
> ACCESS_GRANTED // SECURITY CLEARANCE: VERIFIED PARTICIPANT
OPERATIVE: MUDITHA1
Welcome to the active Kernel0X CTF competition arena. Your credentials have been verified through secure email token handshake. Stage 01 (OSINT) is currently unlocked and live for submission.
CURRENT RANK
#01
STAGE TIMER // XP NOW
—
STAGE STATUS
STAGE 6 ACTIVE
STAGE 01 SOLVED
OSINT
// RECON
STAGE 02 SOLVED
STEGANO
// CARRIER
STAGE 03 SOLVED
CRYPTO
// CIPHER
STAGE 04 SOLVED
NETWORK
// PCAP
STAGE 05 SOLVED
CIPHER II
// VIGENÈRE
STAGE 06 OPEN
LIVE SYS
// CONTAINER
> CATEGORY: CRYPTOGRAPHY
DIFFICULTY: HARD
150 PTS
STAGE 5 — THE SECOND CIPHER
The network evidence has revealed another message left by Kernel0X.
The message recovered from the FTP data channel is encrypted and cannot be read directly. However, the FTP control conversation from the previous stage revealed an important hostname.
Use the information recovered from Stage 4 to determine the key and decrypt the ciphertext.
TOP OPERATIVES LEADERBOARD
LIVE
#1 Mudithal (YOU)
836 XP
5 stages · 21:23:20
#2 muggler
750 XP
5 stages · 00:00
#3 test_op_1791374767048
750 XP
5 stages · 00:00
#4 agent_1791367796779
600 XP
4 stages · 00:00

The decrypted message contains the flag for this stage and information required to continue to the final investigation environment.

You already have the ciphertext from Stage 4. No new file is required.

#5 AAA

493 XP

3 stages · 1:37:02

// CRACK THE CODE — FIND THE VIGENÈRE KEY (9 LETTERS)

KEY DEDUCTION PUZZLE

To uncover the Vigenère cipher key, Kernel0X left this 9-letter deduction puzzle. Compare the guessed words with their position clues to deduce the secret key:

4. CRACK THE CODE — 9 LETTERS

YOUR GUESS	CLUE
H O U S E W A R E	7 letters present, all wrongly placed
S O M E W H E R E	5 letters present, 2 correctly placed
H O U S E S A R E	6 letters present, 1 correctly placed
W A R E S O U T H	7 letters present, 3 correctly placed
S H A R E W O R E	6 letters present, 2 correctly placed

FINAL WORD (9 LETTERS)

_ _ _ _ _ _ _ _ _

[KEY DEDUCTION: Cross-reference with Stage 4 FTP hostname]

// VIGENÈRE CIPHER GUIDELINES:

- Review the hostname discovered during the Stage 4 investigation.
- Determine the appropriate Vigenère key from the hostname.
- Standard Vigenère preserves punctuation, spaces, and numbers.
- Extract the Stage 5 flag and record the terminal credentials for Stage 6.

TEST YOUR 9-LETTER KEY GUESS:

WAREHOUSE

CLEAR

KEY CRACKED: "WAREHOUSE" (9 letters) — Use this key to decrypt the Vigenère ciphertext below!

[COPY KEY]

> RECOVERED MESSAGE

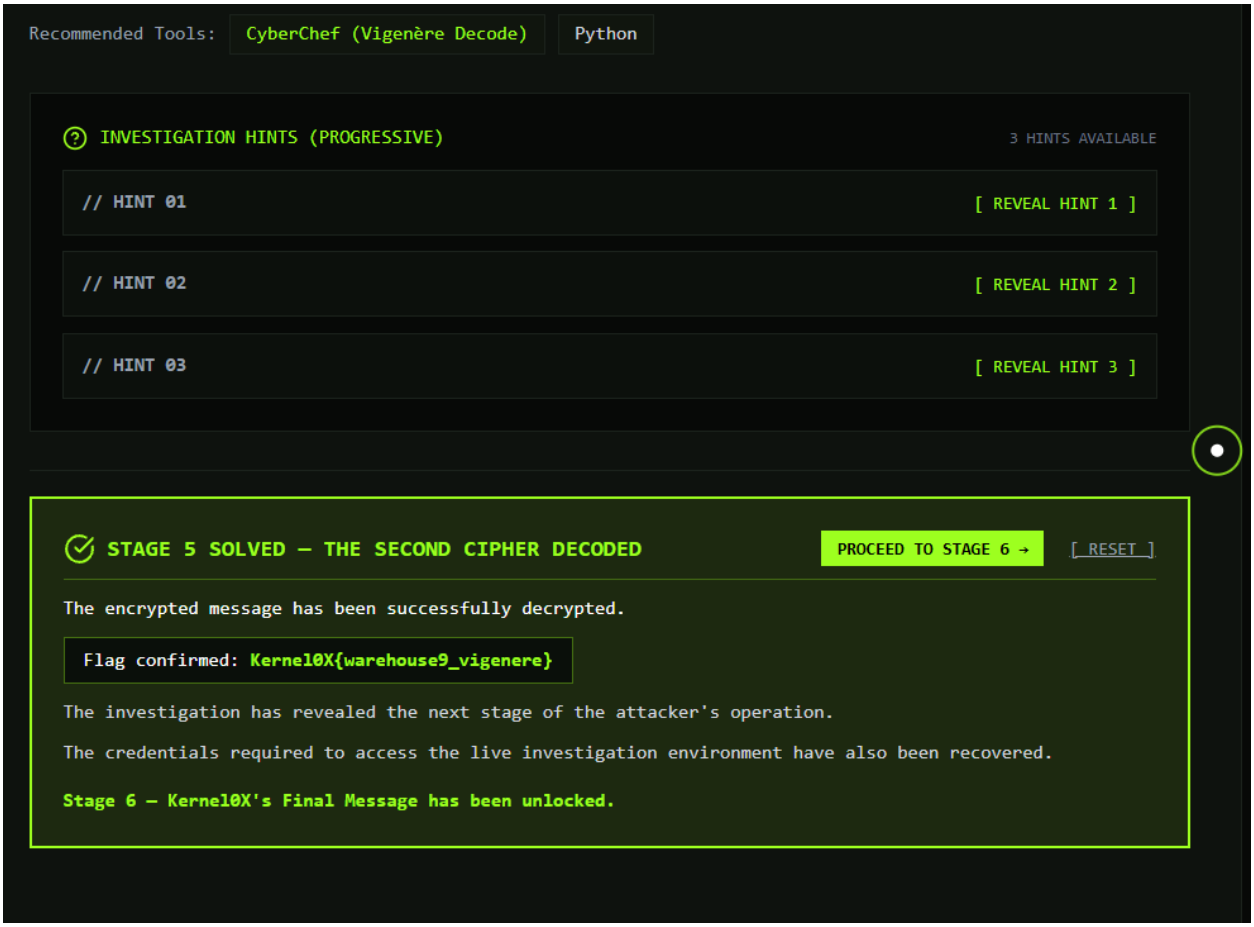
[COPY CIPHERTEXT]

```
OTRKL5_TFSK=Geirlz0R{oeneysbgy9_nmceeiys}|MLECE6_YSZH=mlece6|JXHUY6_HVKTFGVZ=MKL
|OTRKL6_IMWVJADI=r0l_ihinaksy|GNSKA6_PRWZKIJH=Jeoe@2026!
```

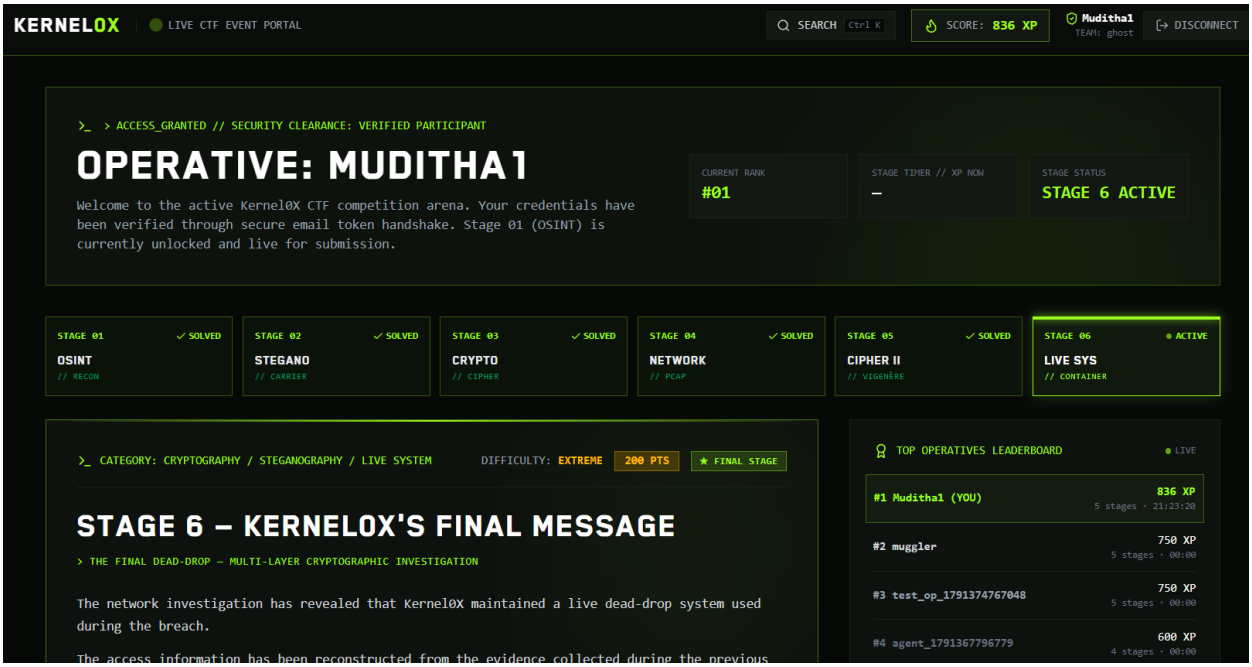
Source: Stage 4 — The Exfiltration Trail

// PLAYER TASK:

1. Review the hostname discovered during the Stage 4 investigation.
2. Determine the appropriate Vigenère key from the hostname.
3. Use the ciphertext recovered in Stage 4.
4. Decrypt the message using the derived key.
5. Recover the Stage 5 flag.
6. Record the credentials revealed in the decrypted message for the final investigation stage.


Recommended Tools:
CyberChef (Vigenère Decode)
Python
INVESTIGATION HINTS (PROGRESSIVE)
3 HINTS AVAILABLE
// HINT 01
[REVEAL HINT 1]
// HINT 02
[REVEAL HINT 2]
// HINT 03
[REVEAL HINT 3]
STAGE 5 SOLVED — THE SECOND CIPHER DECODED
PROCEED TO STAGE 6 →
[RESET]
The encrypted message has been successfully decrypted.
Flag confirmed: Kernel0X{warehouse9_vigenere}
The investigation has revealed the next stage of the attacker's operation.
The credentials required to access the live investigation environment have also been recovered.
Stage 6 — Kernel0X's Final Message has been unlocked.

Stage 6


KERNELOX
LIVE CTF EVENT PORTAL
SEARCH Ctrl K
SCORE: 836 XP
Muditha1
TEAM: ghost
DISCONNECT
> ACCESS_GRANTED // SECURITY CLEARANCE: VERIFIED PARTICIPANT
OPERATIVE: MUDITHA1
Welcome to the active Kernel0X CTF competition arena. Your credentials have been verified through secure email token handshake. Stage 01 (OSINT) is currently unlocked and live for submission.
CURRENT RANK
#01
STAGE TIMER // XP NOW
—
STAGE STATUS
STAGE 6 ACTIVE
STAGE 01 SOLVED
OSINT
// RECON
STAGE 02 SOLVED
STEGANO
// CARRIER
STAGE 03 SOLVED
CRYPTO
// CIPHER
STAGE 04 SOLVED
NETWORK
// PCAP
STAGE 05 SOLVED
CIPHER II
// VIGENÈRE
STAGE 06 ACTIVE
LIVE SYS
// CONTAINER
> CATEGORY: CRYPTOGRAPHY / STEGANOGRAPHY / LIVE SYSTEM
DIFFICULTY: EXTREME
200 PTS
★ FINAL STAGE
STAGE 6 — KERNELOX'S FINAL MESSAGE
> THE FINAL DEAD-DROP — MULTI-LAYER CRYPTOGRAPHIC INVESTIGATION
The network investigation has revealed that Kernel0X maintained a live dead-drop system used during the breach.
The access information has been reconstructed from the evidence collected during the previous
TOP OPERATIVES LEADERBOARD
LIVE
#1 Muditha1 (YOU)
836 XP
5 stages · 21:23:20
#2 muggler
750 XP
5 stages · 00:00
#3 test_op_1791374767048
750 XP
5 stages · 00:00
#4 agent_1791367796779
600 XP
4 stages · 00:00

The access information has been reconstructed from the evidence collected during the previous stages.

You now have a target IP address and port.

Your task is to connect to the restricted server and recover Kernel0X's final message.

The machine contains only the service required for this investigation. No network enumeration is necessary.

The final message is protected by more than one layer. Recover the hidden data, understand the first encrypted message, and use what you discover to solve the second layer.

4 stages · 00:00

#5 AAA

493 XP

3 stages · 1:37:02

// STAGE 6 GUIDELINES:

- Connect via SSH using the recovered credentials.
- Investigate the filesystem thoroughly — not every file is immediately visible.
- The dead-drop image uses a different steganographic technique to Stage 2.
- The first ciphertext requires a keyword-based cipher for decryption.
- The first decryption reveals the key required for the second layer.
- The second ciphertext is in hexadecimal format and uses XOR with a repeating key.
- Decoding both layers correctly reveals the final flag.

>_ > RECOVERED ACCESS CREDENTIALS — FROM PREVIOUS STAGE INVESTIGATION:

TARGET	PORT	PROTOCOL
47.129.24.175	2222	SSH

USERNAME	PASSWORD
Recovered in Stage 1 — Clue / access code discovered during Stage 1 investigation	Recovered in Stage 5 — Your Stage 5 recovered flag, reused as the SSH password

```
ssh -p 2222 <username>@47.129.24.175
```

COPY COMMAND

* Replace <username> with the access code from Stage 1. When prompted for password, enter your Stage 5 flag.

// RECOMMENDED TOOLS:

SSH client · zsteg · CyberChef · Python 3 · xxd / hexdump

> INVESTIGATION OBJECTIVE:

01. Connect to the target server using the credentials above via SSH.
02. Investigate the filesystem. Locate the hidden dead-drop image at: /opt/stage6/dead-drop.png
03. Analyse the image for concealed data. The technique differs from Stage 2 — examine the bit planes and colour channels.
04. Decrypt the first layer of the recovered ciphertext using a repeating keyword-based substitution cipher.
05. Read the first decrypted message carefully — it reveals the key required for the second layer.
06. Locate the hidden dotfile and use the recovered key to decrypt the hexadecimal ciphertext it contains.
07. Submit the final flag using the form below.

INVESTIGATION HINTS (PROGRESSIVE) — 6 HINTS AVAILABLE

// HINT 01 [HIDE HINT]

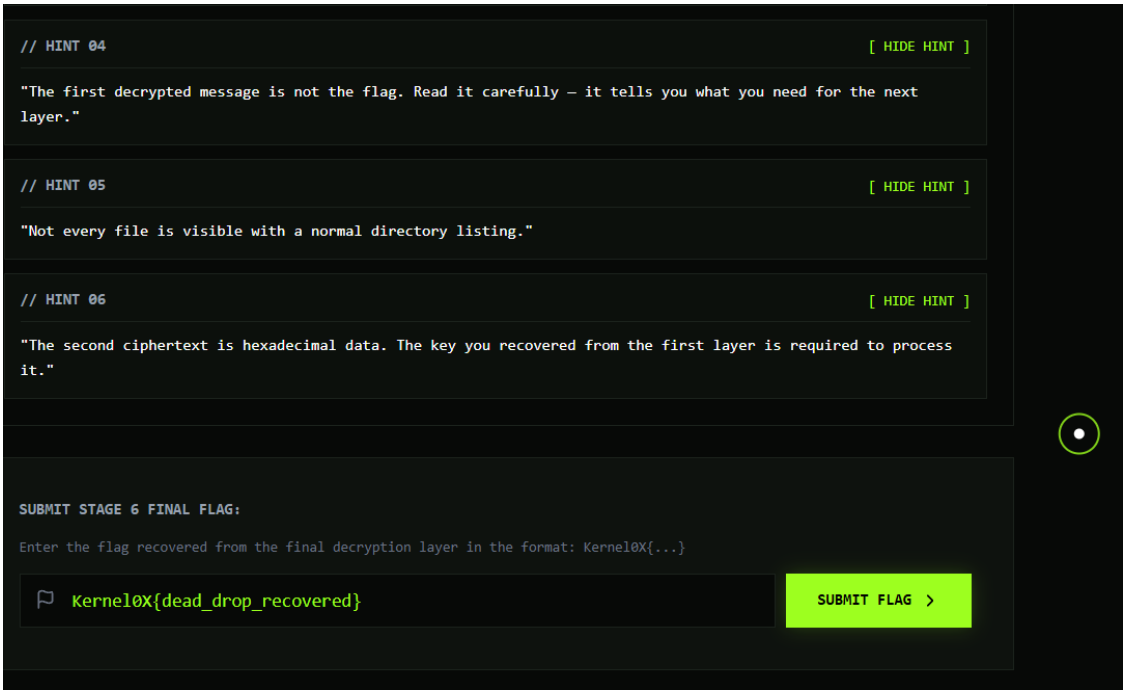
"You already recovered two pieces of information during the investigation. One identifies the account; the other authenticates it."

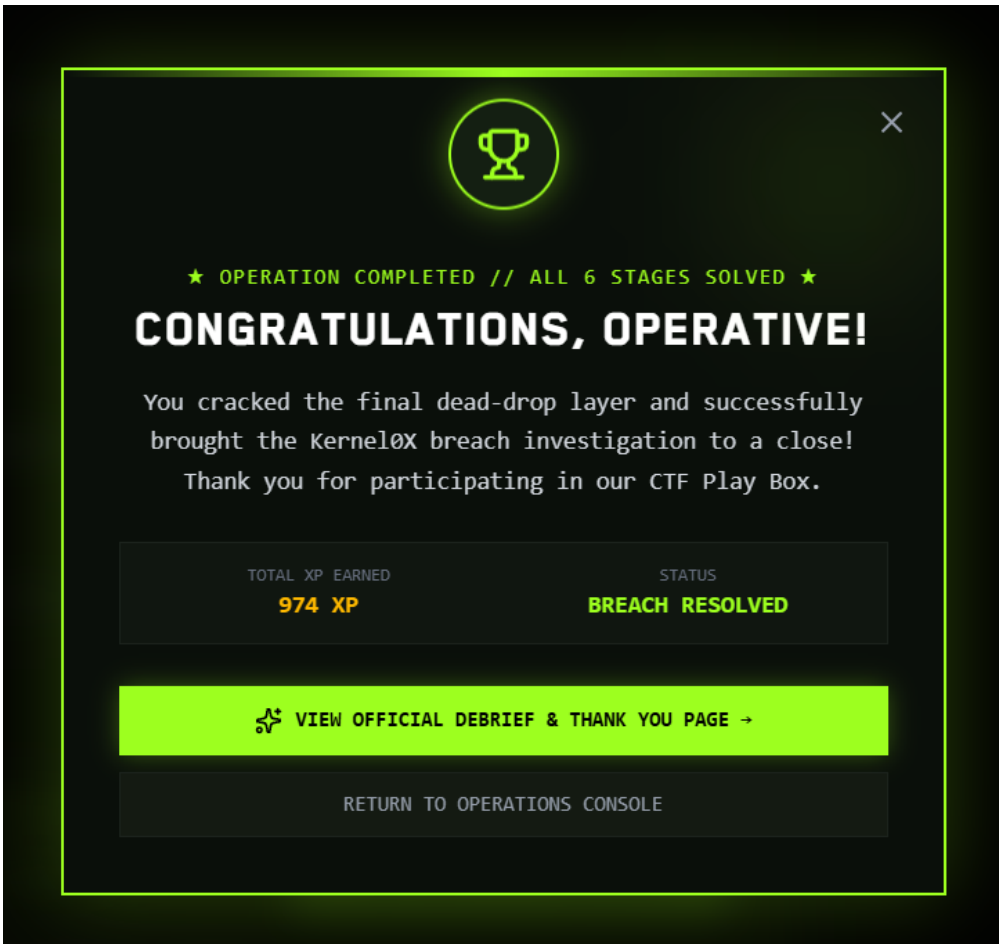
// HINT 02 [HIDE HINT]

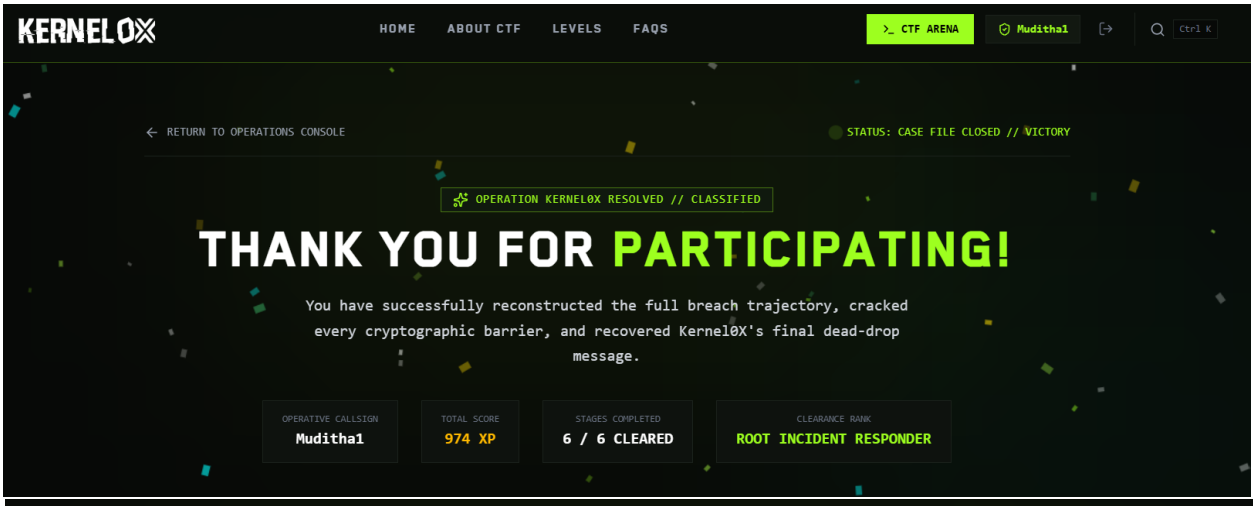
"This image does not rely on the same hiding technique used earlier. Examine its individual bit planes and colour channels."

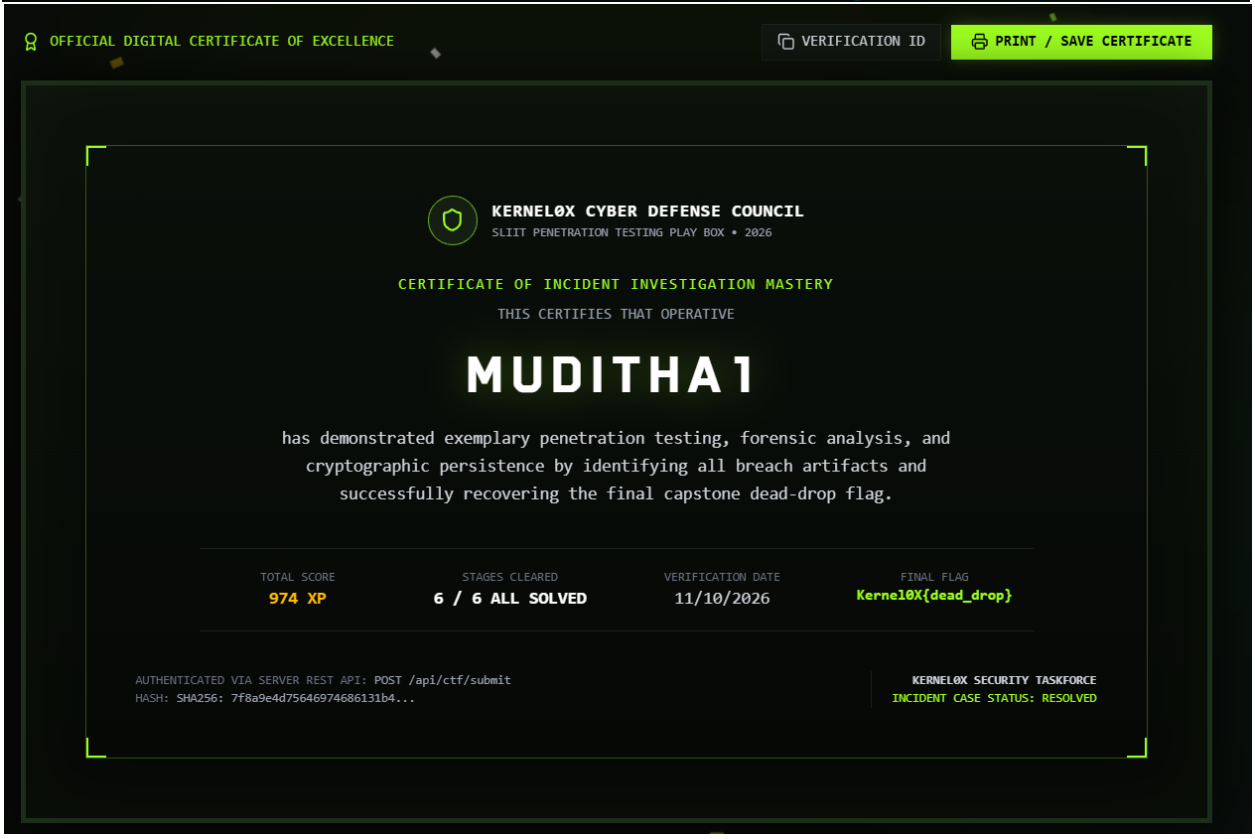
// HINT 03 [HIDE HINT]

"The recovered text is structured ciphertext. A repeating keyword-based substitution cipher may be useful."


// HINT 04
[HIDE HINT]
"The first decrypted message is not the flag. Read it carefully — it tells you what you need for the next layer."
// HINT 05
[HIDE HINT]
"Not every file is visible with a normal directory listing."
// HINT 06
[HIDE HINT]
"The second ciphertext is hexadecimal data. The key you recovered from the first layer is required to process it."
SUBMIT STAGE 6 FINAL FLAG:
Enter the flag recovered from the final decryption layer in the format: Kernel0X{...}
Kernel0X{dead_drop_recovered}
SUBMIT FLAG


★ OPERATION COMPLETED // ALL 6 STAGES SOLVED ★
CONGRATULATIONS, OPERATIVE!
You cracked the final dead-drop layer and successfully brought the Kernel0X breach investigation to a close! Thank you for participating in our CTF Play Box.
TOTAL XP EARNED
974 XP
STATUS
BREACH RESOLVED
VIEW OFFICIAL DEBRIEF & THANK YOU PAGE →
RETURN TO OPERATIONS CONSOLE


KERNEL0X
HOME
ABOUT CTF
LEVELS
FAQS
CTF ARENA
Muditha1
Ctrl K
RETURN TO OPERATIONS CONSOLE
STATUS: CASE FILE CLOSED // VICTORY
OPERATION KERNEL0X RESOLVED // CLASSIFIED
THANK YOU FOR PARTICIPATING!
You have successfully reconstructed the full breach trajectory, cracked every cryptographic barrier, and recovered Kernel0X's final dead-drop message.
OPERATIVE CALLSIGN
Muditha1
TOTAL SCORE
974 XP
STAGES COMPLETED
6 / 6 CLEARED
CLEARANCE RANK
ROOT INCIDENT RESPONDER


OFFICIAL DIGITAL CERTIFICATE OF EXCELLENCE
VERIFICATION ID
PRINT / SAVE CERTIFICATE
KERNEL0X CYBER DEFENSE COUNCIL
SLIIT PENETRATION TESTING PLAY BOX • 2026
CERTIFICATE OF INCIDENT INVESTIGATION MASTERY
THIS CERTIFIES THAT OPERATIVE
MUDITHA1
has demonstrated exemplary penetration testing, forensic analysis, and cryptographic persistence by identifying all breach artifacts and successfully recovering the final capstone dead-drop flag.
TOTAL SCORE
974 XP
STAGES CLEARED
6 / 6 ALL SOLVED
VERIFICATION DATE
11/10/2026
FINAL FLAG
Kernel0X{dead_drop}
AUTHENTICATED VIA SERVER REST API: POST /api/ctf/submit
HASH: SHA256: 7f8a9e4d75646974686131b4...
KERNEL0X SECURITY TASKFORCE
INCIDENT CASE STATUS: RESOLVED

Testing evidence

```
jemoirs@ghost:~/Kernel0X/Kernel0x$ git --version
git version 2.47.3
jemoirs@ghost:~/Kernel0X/Kernel0x$ node -version
node: bad option: -version
jemoirs@ghost:~/Kernel0X/Kernel0x$ node --version
v20.19.2
jemoirs@ghost:~/Kernel0X/Kernel0x$ npm --version
9.2.0
jemoirs@ghost:~/Kernel0X/Kernel0x$ git --version
git version 2.47.3
jemoirs@ghost:~/Kernel0X/Kernel0x$ node --version
v20.19.2
jemoirs@ghost:~/Kernel0X/Kernel0x$ npm --version
9.2.0
jemoirs@ghost:~/Kernel0X/Kernel0x$ python3 --version
Python 3.13.5
```

```
jemoirs@ghost:~/Kernel0X/Kernel0x$ ls -la
total 148
drwxrwxr-x  8 jemoirs jemoirs  4096 Oct 11 19:10 .
drwxrwxr-x  3 jemoirs jemoirs  4096 Oct 11 19:08 ..
drwxrwxr-x  5 jemoirs jemoirs  4096 Oct 11 19:30 backend
-rw-r--r--  1 jemoirs jemoirs 21137 Oct 11 15:32 dead-drop.png
-rw-rw-r--  1 jemoirs jemoirs  5897 Oct 11 12:11 deploy_ctf_platform.py
drwxrwxr-x  4 jemoirs jemoirs  4096 Oct 11 20:07 evidence
drwxrwxr-x  5 jemoirs jemoirs  4096 Oct 11 12:18 frontend
drwxrwxr-x  8 jemoirs jemoirs  4096 Oct 11 12:12 .git
-rw-rw-r--  1 jemoirs jemoirs    74 Oct 11 12:11 .gitignore
-rw-rw-r--  1 jemoirs jemoirs  4059 Oct 11 12:11 member2_stages_solver.py
-rw-rw-r--  1 jemoirs jemoirs  4771 Oct 11 12:11 member3_stages_solver.py
drwxrwxr-x 27 jemoirs jemoirs  4096 Oct 11 19:08 node_modules
-rw-rw-r--  1 jemoirs jemoirs   430 Oct 11 12:11 package.json
-rw-rw-r--  1 jemoirs jemoirs 10579 Oct 11 19:08 package-lock.json
drwxrwxr-x  3 jemoirs jemoirs  4096 Oct 11 12:11 public
-rw-rw-r--  1 jemoirs jemoirs   974 Oct 11 12:11 README.md
-rw-rw-r--  1 jemoirs jemoirs  2855 Oct 11 12:11 stage1_osint_extract.py
-rw-rw-r--  1 jemoirs jemoirs  2491 Oct 11 12:11 stage2_stego_extract.py
-rw-rw-r--  1 jemoirs jemoirs  2096 Oct 11 12:11 stage3_caesar_solver.py
-rw-rw-r--  1 jemoirs jemoirs  5272 Oct 11 12:11 stage4_pcap_analyzer.py
-rw-rw-r--  1 jemoirs jemoirs   679 Oct 11 17:39 stage5.py
-rw-rw-r--  1 jemoirs jemoirs  4023 Oct 11 12:11 stage5_vigenere_solver.py
-rw-rw-r--  1 jemoirs jemoirs   566 Oct 11 15:36 stage6_1.py
-rw-rw-r--  1 jemoirs jemoirs   528 Oct 11 15:39 stage6_2.py
-rw-rw-r--  1 jemoirs jemoirs  3859 Oct 11 12:11 stage6_capstone_solver.py
-rw-rw-r--  1 jemoirs jemoirs  7923 Oct 11 12:11 verify_e2e_pipeline.py
```

```
jemoirs@ghost:~/Kernel0X/Kernel0x/frontend$ npm install

up to date in 611ms

132 packages are looking for funding
  run `npm fund` for details
jemoirs@ghost:~/Kernel0X/Kernel0x/frontend$ npm run
Scripts available in frontend@0.0.0 via `npm run-script`:
  dev
    vite
  build
    vite build
  lint
    oxlint
  preview
    vite preview

jemoirs@ghost:~/Kernel0X/Kernel0x/frontend$ npm run dev

> frontend@0.0.0 dev
> vite

Port 5173 is in use, trying another one...

  VITE v8.2.2  ready in 386 ms

  ➜  Local:   http://localhost:5174/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
```

```
jemoirs@ghost:~/Kernel0X/Kernel0x/backend$ npm run
Lifecycle scripts included in kernel0x-backend@1.0.0:
  start
    node src/server.js

available via `npm run-script`:
  dev
    node --watch src/server.js

jemoirs@ghost:~/Kernel0X/Kernel0x/backend$ npm install

up to date in 305ms

18 packages are looking for funding
  run `npm fund` for details
jemoirs@ghost:~/Kernel0X/Kernel0x/backend$ npm run
Lifecycle scripts included in kernel0x-backend@1.0.0:
  start
    node src/server.js

available via `npm run-script`:
  dev
    node --watch src/server.js

jemoirs@ghost:~/Kernel0X/Kernel0x/backend$
```

```
jemoirs@ghost:~/Kernel0X/Kernel0x$ find . -maxdepth 3 -type f \( -name 'verify_e2e_pipeline.py' -o -name 'package.json' -o -iname '*test*' -o -iname '*readme*' \)
./evidence/member4-testing/e2e-test.log
./node_modules/ansi-regex/readme.md
./node_modules/ansi-regex/package.json
./node_modules/tslib/README.md
./node_modules/tslib/package.json
./node_modules/wrap-ansi/readme.md
./node_modules/wrap-ansi/package.json
./node_modules/yargs/README.md
./node_modules/yargs/package.json
./node_modules/require-directory/package.json
./node_modules/require-directory/README.markdown
./node_modules/string-width/readme.md
./node_modules/string-width/package.json
./node_modules/cliui/README.md
./node_modules/cliui/package.json
./node_modules/ansi-styles/readme.md
./node_modules/ansi-styles/package.json
./node_modules/shell-quote/README.md
./node_modules/shell-quote/package.json
./node_modules/y18n/README.md
./node_modules/y18n/package.json
./node_modules/rxjs/README.md
./node_modules/rxjs/package.json
./node_modules/emoji-regex/README.md
./node_modules/emoji-regex/package.json
./node_modules/supports-color/readme.md
./node_modules/supports-color/package.json
./node_modules/get-caller-file/README.md
./node_modules/get-caller-file/package.json
./node_modules/is-fullwidth-code-point/readme.md
./node_modules/is-fullwidth-code-point/package.json
./node_modules/escalade/readme.md
./node_modules/escalade/package.json
./node_modules/strip-ansi/readme.md
./node_modules/strip-ansi/package.json
./node_modules/chalk/readme.md
./node_modules/chalk/package.json
./node_modules/concurrently/README.md
./node_modules/concurrently/package.json
./node_modules/tree-kill/README.md
```

```
jemoirs@ghost:~/Kernel0X/Kernel0x$ python3 verify_e2e_pipeline.py

================================================================================
Kernel0X — End-to-End Integration & Verification Suite (Member 4)
================================================================================
Target API: http://localhost:5000/api  |  AWS Sandbox: 47.129.24.175:2222

TC-01    | Backend Gateway Health & Liveness Audit             | [PASS] | Status: Online (200)
TC-02    | Automated Operative Registration & JWT Issuance     | [PASS] | User: auditor_1791733031
TC-03    | Sequential Dependency Lock (Stage 2 jump blocked)   | [PASS] | HTTP 403 Forbidden
TC-04    | Malformed / Invalid Flag Rejection                  | [PASS] | HTTP 400 Bad Request
TC-05    | Stage 1 Solve: OSINT Metadata Clue (K0X-17)         | [PASS] | +200 XP
TC-06    | Stage 2 Solve: Steghide DCT Payload Extraction      | [PASS] | +200 XP
TC-07    | Stage 3 Solve: Caesar ROT-7 Flag Decryption         | [PASS] | +200 XP
TC-08    | Stage 4 Solve: PCAP FTP Stream Exfiltration         | [PASS] | Bridge: Ciphertext forwarded
TC-09    | Stage 5 Solve: Vigenere Keyword Decrypt (WAREHOUSE) | [PASS] | Credentials derived
TC-10    | Stage 6 Solve: Capstone SSH + LSB + XOR Flag        | [PASS] | Total XP: 1200
TC-11    | Platform Recovery: Full Reset & State Purge         | [PASS] | Clean State Restored
TC-12    | Security Isolation: AWS EC2 SSH Port 2222 Liveness  | [PASS] | 47.129.24.175:2222 Open

--------------------------------------------------------------------------------
Verification Summary: 12/12 Test Cases Passed (100.0%)
Status: ALL INTEGRATION, DEPENDENCY & RESET TESTS PASSED SUCCESSFULLY!
================================================================================
```

```
jemoirs@ghost:~/Kernel0X/Kernel0x$ mkdir -p evidenve/member4-testing
jemoirs@ghost:~/Kernel0X/Kernel0x$ python3 verify_e2e_pipeline.py   2>&1 | tee evidence/member4-testing/e2e-test.log

================================================================================
  Kernel0X — End-to-End Integration & Verification Suite (Member 4)
================================================================================
  Target API: http://localhost:5000/api  |  AWS Sandbox: 47.129.24.175:2222

  TC-01    | Backend Gateway Health & Liveness Audit            | [PASS] | Status: Online (200)
  TC-02    | Automated Operative Registration & JWT Issuance    | [PASS] | User: auditor_1791733101
  TC-03    | Sequential Dependency Lock (Stage 2 jump blocked)  | [PASS] | HTTP 403 Forbidden
  TC-04    | Malformed / Invalid Flag Rejection                 | [PASS] | HTTP 400 Bad Request
  TC-05    | Stage 1 Solve: OSINT Metadata Clue (K0X-17)        | [PASS] | +200 XP
  TC-06    | Stage 2 Solve: Steghide DCT Payload Extraction     | [PASS] | +200 XP
  TC-07    | Stage 3 Solve: Caesar ROT-7 Flag Decryption        | [PASS] | +200 XP
  TC-08    | Stage 4 Solve: PCAP FTP Stream Exfiltration        | [PASS] | Bridge: Ciphertext forwarded
  TC-09    | Stage 5 Solve: Vigenere Keyword Decrypt (WAREHOUSE)| [PASS] | Credentials derived
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  Verification Summary: 12/12 Test Cases Passed (100.0%)
  Status: ALL INTEGRATION, DEPENDENCY & RESET TESTS PASSED SUCCESSFULLY!
================================================================================

jemoirs@ghost:~/Kernel0X/Kernel0x$ find . -maxdepth 3 -type f   \( -name 'verify_e2e_pipeline.py'   -o -name 'package.json'   -o -iname '*test*'   -o -iname '*readme*' \)
./evidence/member4-testing/e2e-test.log
```

```
jemoirs@ghost:~/Kernel0X/Kernel0x$ find . -maxdepth 3 -type f   \( -name 'verify_e2e_pipeline.py'   -o -name 'package.json'   -o -iname '*test*'   -o -iname '*readme*' \)
./node_modules/tslib/README.md
./node_modules/tslib/package.json
./node_modules/wrap-ansi/readme.md
./node_modules/wrap-ansi/package.json
./node_modules/yargs/README.md
./node_modules/yargs/package.json
./node_modules/require-directory/package.json
./node_modules/require-directory/README.markdown
./node_modules/string-width/readme.md
./node_modules/string-width/package.json
./node_modules/cliui/README.md
./node_modules/cliui/package.json
./node_modules/ansi-styles/readme.md
./node_modules/ansi-styles/package.json
./node_modules/shell-quote/README.md
./node_modules/shell-quote/package.json
./node_modules/y18n/README.md
./node_modules/y18n/package.json
./node_modules/rxjs/README.md
./node_modules/rxjs/package.json
./node_modules/emoji-regex/README.md
./node_modules/emoji-regex/package.json
./node_modules/supports-color/readme.md
./node_modules/supports-color/package.json
./node_modules/get-caller-file/README.md
./node_modules/get-caller-file/package.json
./node_modules/is-fullwidth-code-point/readme.md
./node_modules/is-fullwidth-code-point/package.json
./node_modules/escalade/readme.md
./node_modules/escalade/package.json
./node_modules/strip-ansi/readme.md
./node_modules/strip-ansi/package.json
./node_modules/chalk/readme.md
./node_modules/chalk/package.json
./node_modules/concurrently/README.md
./node_modules/concurrently/package.json
./node_modules/tree-kill/README.md
./node_modules/tree-kill/package.json
./node_modules/has-flag/readme.md
./node_modules/has-flag/package.json
./node_modules/color-name/README.md
./node_modules/color-name/package.json
./node_modules/yargs-parser/README.md
./node_modules/yargs-parser/package.json
./node_modules/color-convert/README.md
./node_modules/color-convert/package.json
./verify_e2e_pipeline.py
./frontend/README.md
./frontend/package.json
./README.md
./package.json
./backend/package.json
```

